

AhamX Bodhi

Where I Becomes Infinite

Progressive Skill Stack

Module 1 · Course Learning Path
Prof. Sashikumaar Ganesan · IISc Bengaluru



Session Overview

This Session

- What is a Progressive Skill Stack?
- The five skill tiers in GenAI
- How skills build on each other
- Your position in the stack today

Foundation Required

- Basic Python programming
- Comfort with command-line tools
- Understanding of what an LLM is
- Module 1.1 and 1.2 completed

Learning Objectives



By the End of This Session You Will Be Able To

1. **Describe** the five tiers of the Progressive Skill Stack for GenAI
2. **Explain** how each tier builds upon the previous one
3. **Identify** your current position within the skill stack
4. **Connect** the skill stack tiers to the course module sequence
5. **Articulate** the learning pathway from user to innovator



What is a Progressive Skill Stack?

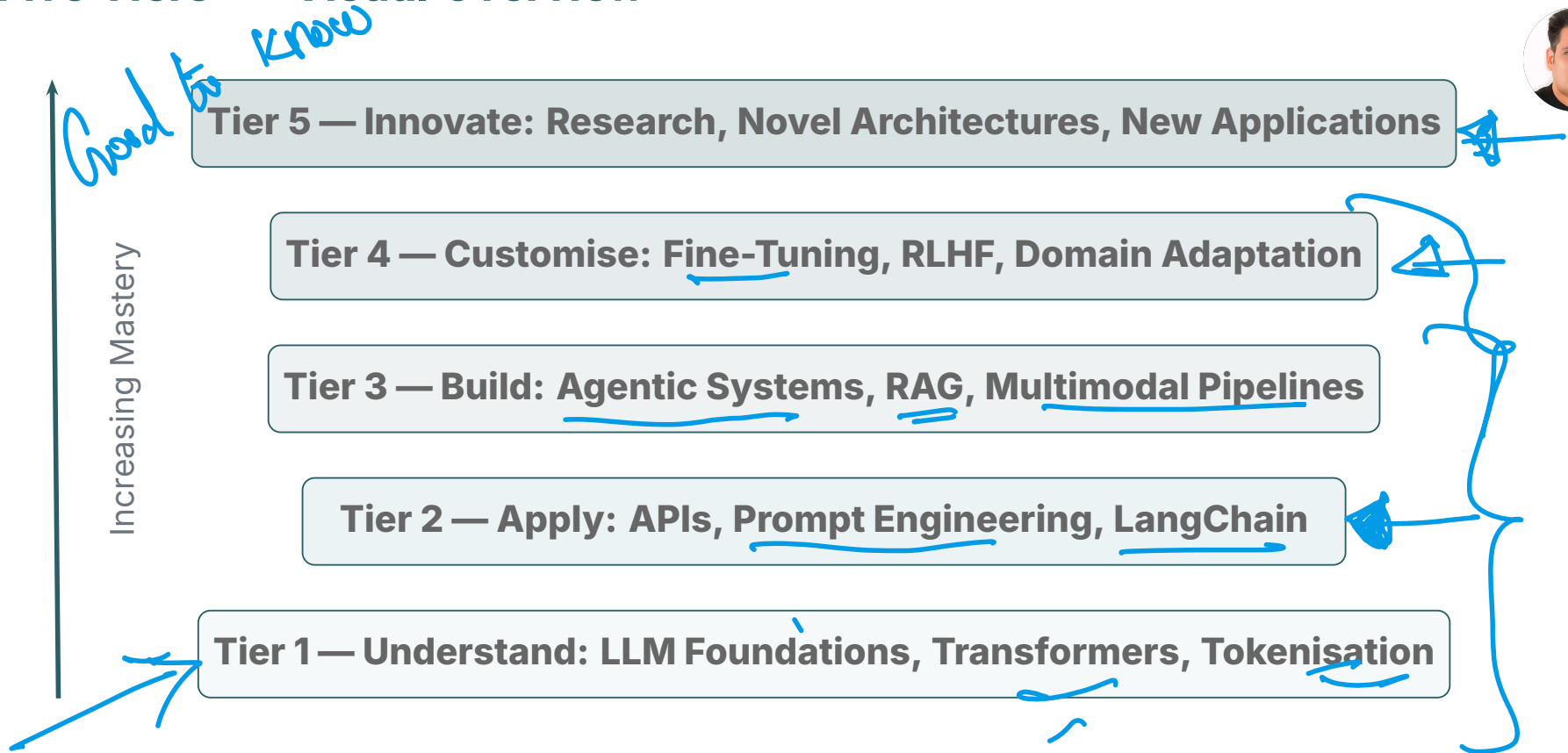
Definition

A **Progressive Skill Stack** is a structured, layered framework in which each level of competency builds directly upon mastery of the level below it. In Generative AI, skills are organized so that foundational abilities enable intermediate ones, which in turn unlock advanced capabilities.

- Skills are **cumulative** — earlier layers are never discarded
- Each tier has its own **tools, vocabulary, and deliverables**
- The stack maps directly to this course's **module sequence**
- Progress is measured by what you can **build and deploy**, not just understand



The Five Tiers --- Visual Overview





Tier 1 --- Understand: LLM Foundations

What You Learn

- How language models work internally
- Transformer architecture and self-attention
- Tokenisation and vocabulary construction
- Autoregressive text generation
- Foundation model landscape (GPT, Llama)

Course Modules

- Module 1: GenAI Foundations
- Module 2: Core Architecture
- Module 3: Large Language Models

Deliverable

- Explain how an LLM generates a token
- Read and discuss architecture papers



Tier 2 --- Apply: APIs and Prompt Engineering

What You Learn

- Calling LLM APIs (OpenAI, Anthropic, Hugging Face)
- Temperature, Top-P, Top-K parameter tuning
- Zero-shot, few-shot, chain-of-thought prompting
- Building prompt templates and chains
- LangChain for orchestration

Course Modules

- Module 2: Prompt Engineering sections
- Module 3: API Integration labs

Deliverable

- Production-quality prompt templates
- Working API integration with tuned parameters



Tier 3 --- Build: Agentic and Multimodal Systems

What You Learn

- Retrieval-Augmented Generation (RAG) pipelines
- AI agent design: ReAct, planning, tool use
- Multi-agent coordination (LangGraph, MCP)
- Vision-Language and Audio-Speech models
- Context management and vector databases

Course Modules

- Module 4: Multimodal AI Systems
- Module 5: Introduction to Agentic AI

Deliverable

- End-to-end RAG agent
- Multimodal application prototype



Tier 4 --- Customise: Fine-Tuning and Alignment

What You Learn

- Full fine-tuning vs. parameter-efficient fine-tuning
- LoRA, QLoRA, and adapter modules
- Instruction-following dataset construction
- RLHF: reward models and PPO optimisation
- Evaluation: benchmarks and LLM-as-judge

Course Modules

- Module 7: Foundational Model Fine-tuning
- Module 8: AI Safety and Guardrails

Deliverable

- Domain-adapted fine-tuned model
- Safety evaluation framework



Tier 5 --- Innovate: Research and Novel Systems

What You Learn

- Pre-training from scratch with custom corpora
- LLMOps: serving, monitoring, scaling
- Federated and distributed training
- Scientific ML and Physics-Informed AI
- Production GenAI system integration

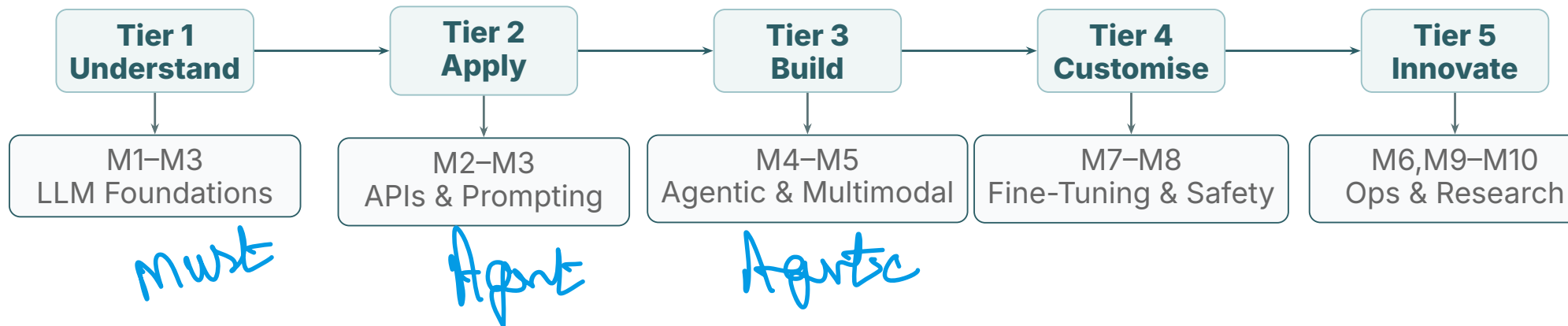
Course Modules

- Module 6: Scientific ML
- Module 9: LLMOps
- Module 10: Federated AI
- Capstone Project

Deliverable

- Production-grade AI system
- Capstone demonstration

Skill Stack --- Course Module Mapping





Why a Stack? The Dependency Logic

Skills Are Not Independent

Each tier requires specific competencies from the tier below it. Attempting Tier 3 (Build) without Tier 2 (Apply) means working with APIs and prompt patterns you have not yet internalised — leading to brittle systems and slow debugging.

- **Tier 2 requires Tier 1:** You cannot tune temperature without understanding how tokens are sampled
- **Tier 3 requires Tier 2:** RAG pipelines call LLM APIs inside retrieval loops
- **Tier 4 requires Tier 3:** Fine-tuning objectives build on inference-time behaviour understanding
- **Tier 5 requires Tier 4:** Pre-training at scale requires mastery of all adaptation techniques



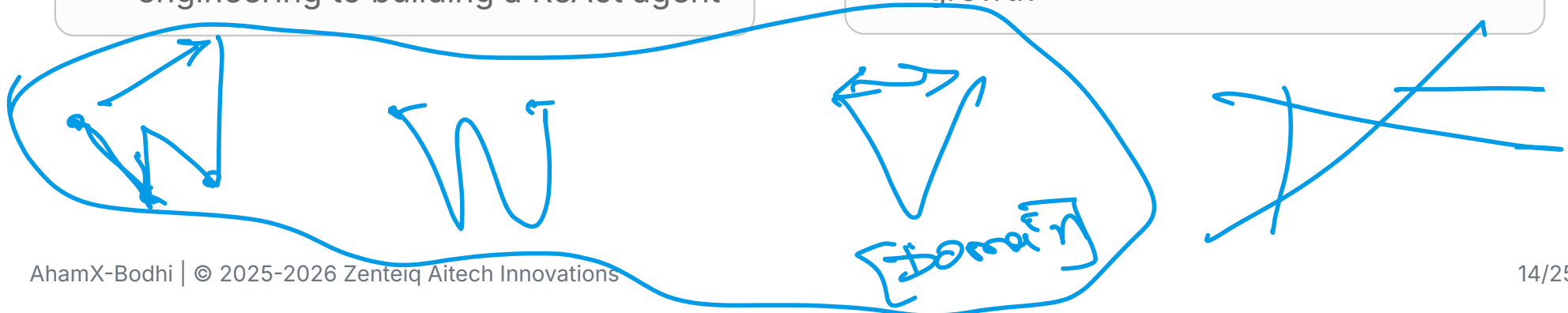
Horizontal vs. Vertical Growth in the Stack

Vertical Growth (Depth)

- Moving **up** the stack to the next tier
- Acquiring new conceptual frameworks
- Requires completing prerequisites
- Measured by: new deliverables produced
- Example: moving from prompt engineering to building a ReAct agent

Horizontal Growth (Breadth)

- Expanding coverage **within** a tier
- Applying the tier to new domains
- Example: applying Tier 2 skills to audio and vision models
- Both types of growth are needed for professional practice
- Module mini-projects train horizontal growth





Application: Enterprise Customer Support AI

How the Stack Maps to a Real System

A production customer-support chatbot at a bank requires **all five tiers**:

- **Tier 1:** Engineers must understand why the LLM hallucinates account details
- **Tier 2:** Prompt templates enforce response format; API parameters control verbosity
- **Tier 3:** RAG retrieves live policy documents; agents escalate to human agents
- **Tier 4:** Fine-tuning on internal ticket data improves domain accuracy
- **Tier 5:** LLMOps monitors latency, drift, and compliance at scale

Key Insight

No single tier is sufficient for production. The stack is the product.



Application: AI-Assisted Medical Diagnostics

Stack in Clinical NLP

A radiology report summarisation tool illustrates progressive skill dependency:

- **Tier 1:** Understanding attention patterns helps explain model focus on findings
- **Tier 2:** Structured prompts extract DICOM metadata reliably
- **Tier 3:** Multimodal pipelines process both images and report text jointly
- **Tier 4:** Fine-tuning on de-identified clinical notes improves terminology accuracy
- **Tier 5:** Federated training (Module 10) preserves patient privacy across hospitals

Regulatory Implication

Each tier adds an auditable layer, which matters for FDA/CE compliance.



Application: Physics-Informed AI for Engineering

Where Scientific ML Sits in the Stack

Physics-Informed Neural Networks (PINNs) occupy the **Tier 5 Innovate** layer:

- Requires Tier 1 understanding of neural network loss landscapes
- Requires Tier 3 pipeline skills to integrate PDE solvers with training loops
- Tier 4 domain adaptation tunes PINN loss weights for specific geometries
- Tier 5 LLMOps manages large-scale surrogate model serving

Module 6 Connection

Scientific ML and PINNs (Module 6) are a direct application of the full skill stack to computational science and engineering domains.



Application: AI-Powered Code Generation Tools

Copilot-Style Products

- **Tier 1:** Know why LLMs predict next tokens in code context
- **Tier 2:** Prompt design for language, framework, style constraints
- **Tier 3:** Agentic code review and test generation loops
- **Tier 4:** Fine-tune on proprietary codebase style guides
- **Tier 5:** Pre-train on internal repositories; deploy with LLMOps

Industry Examples

- GitHub Copilot (OpenAI Codex / GPT-4)
- Amazon CodeWhisperer
- Google Gemini Code Assist
- Tabnine (on-premise fine-tuning)
- Cursor (agentic coding agents)

Where Are You in the Stack Today?



Tier	Entry Criteria	You Can Already...	Target
Tier 1	Completed M1–M2	Explain tokens, transformers	By end of M3
Tier 2	Completed M2–M3	Call APIs, write prompt templates	By end of M3
Tier 3	Completed M4–M5	Build RAG agents, use MCP	By end of M5
Tier 4	Completed M7–M8	Fine-tune LoRA models	By end of M8
Tier 5	Completed M9–M10	Deploy production LLM systems	Capstone



Common Misconceptions About the Skill Stack

Misconceptions to Avoid

- **"I can skip Tier 1 because I use ChatGPT already"** — Using a model is not the same as understanding it; API failures and prompt failures require Tier 1 diagnosis skills
- **"Fine-tuning is always better than prompting"** — Tier 4 customisation is expensive; skilled Tier 2 prompting often achieves equivalent results at a fraction of the cost
- **"Agentic AI replaces prompt engineering"** — Agents are built on top of prompts; Tier 3 requires mastery of Tier 2
- **"The stack is linear — one tier at a time"** — In practice, you work across multiple adjacent tiers simultaneously; the stack is a *mental model*, not a strict gate



Skills You Develop Across the Stack

Technical Skills

- Transformer implementation (PyTorch)
- API integration and parameter tuning
- Vector database querying (Pinecone, Qdrant)
- LoRA/QLoRA fine-tuning pipelines
- Production deployment (FastAPI, Docker, vLLM)

Conceptual & Applied Skills

- Architectural reasoning about model behaviour
- Prompt design for reliability and safety
- Agent evaluation and failure analysis
- Alignment and bias assessment
- Research literature reading and critique

Professional Roles Enabled by Each Tier



Tier	Primary Roles	Sample Job Titles
Tier 1-2	LLM Consumer, Prompt Engineer	AI Analyst, Prompt Specialist
Tier 3	AI Application Developer	AI Engineer, Solutions Architect
Tier 4	Model Adaptation Specialist	<u>ML Engineer</u> , <u>NLP Engineer</u>
<u>Tier 5</u>	AI Research and Infra Engineer	Research <u>Scientist</u> , <u>MLOps</u> Engineer
<u>All Tiers</u>	GenAI Product Leadership	<u>AI Product Manager</u> , <u>CTO</u>



Knowledge Graph --- Prerequisites and Connections

Requires (Prerequisites)

- Five Level Framework for GenAI (AI-GN-FN-TH-000006)
- Foundation Models Overview (AI-LM-FN-TH-000005)
- What is Generative AI? (AI-GN-FN-TH-000001)

This Concept Unlocks

- From Using to Building Models (AI-GN-FN-TH-000008)
- Module 2 Five Level Framework content
- All subsequent skill-building modules (M2-Capstone)

Summary



Key Takeaways from This Session

1. The **Progressive Skill Stack** organises GenAI competencies into five cumulative tiers: Understand, Apply, Build, Customise, Innovate
2. Each tier has a direct mapping to **course modules**, tools, and deliverables
3. Skills are **cumulative** — earlier tiers are prerequisites for, not replacements of, later ones
4. Both **vertical** (depth) and **horizontal** (breadth) growth are required for professional practice
5. The stack is a **mental model for self-assessment**: it shows where you are and what to target next

Thank You

Progressive Skill Stack · Module 1.3 · AI-GN-FN-TH-000007